TÉCNICO
UNIVERSIDADE
DE LISBOA

SHIP TYPES 1.1 · 10 MIN

Classification of Ship Types

Main types of ships: **cargo carrying** (bulk carriers, dry cargo and passenger ships, tankers) · **work vessels** (fishing, offshore, dredging, tugs, service craft) · **non-merchant ships** (naval, yachts, training) · inland, non-propelled and non-ship units

Session W1.1 · Not in the Evaluation Test

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“There are no universally applicable definitions of ship types.”

IMO, Safety Regulations, imo.org

Despite the lack of cross-framework consistency, specific names and definitions are used rigorously inside each framework. IMO conventions, EU MRV and FuelEU Maritime, classification societies and commercial data providers each have their own taxonomy. Each was designed for a different purpose (regulatory compliance, commercial data, technical class rules, operational training) and each contributes something the others miss.

THE THREE KEY FRAMEWORKS

IMO conventions

Regulatory compliance

SOLAS, MARPOL and the codes; MARPOL Annex VI sets the types for EEDI, EEXI and CII

EU MRV, ETS and FuelEU

EU reporting and compliance

One list of ship types, each with its cargo measure, reused by all three

StatCode 5

Commercial data and statistics

S&P Global’s five-level coding, used by ship databases and statistics offices

Six Families of Ships, Coded in Five Levels

One ship, five levels

Level 1 · Family
Cargo carrying

Level 2 · Group
Dry cargo/passenger

Level 3 · Sub-group
Container

Level 4 · Ship type
Container ship

Level 5 · Detailed type
Container ship (fully cellular)

Cargo carrying

Bulk carriers · Dry cargo/passenger · Tankers

Our two ships and the reference ship

Work vessel

Fishing · Offshore · Miscellaneous (dredging, research, towing, other activities)

Non merchant

Naval and naval auxiliaries · Yachts · Sail training ships

Non seagoing merchant

Inland waterway cargo ships, tankers, passenger ships, tugs and dredgers

Non propelled

Barges and pontoons

Non ship structure

Platforms, floating docks, buoys, submersibles, hovercraft

The World Fleet in 2025: Capacity and Growth

1 January 2025 · **112,501 ships** of 100 GT and above · **2.44 billion dwt**, 3.4 % more than a year earlier

Type	Share of world deadweight	Share	Growth 2024
Bulk carriers		42.5 %	+3.0 %
Oil tankers		27.5 %	+0.7 %
Container ships		**14.9 %**	**+9.7 %**
Liquefied gas carriers		4.1 %	+7.0 %
Offshore supply		3.7 %	+2.0 %
General cargo		3.6 %	+1.9 %
Chemical tankers		2.2 %	+3.3 %
Other and unknown		1.1 %	+4.7 %
Ferries and passenger ships		0.4 %	+1.5 %

Container ships grew fastest, by **9.7 %** in one year. The fleet is also getting older: **12.6 years** on average by tonnage, **22.2 years** by number of ships.

Who Owns, Flags, Builds and Recycles the Fleet

Who owns it

Share of world deadweight, by owner

16.4 % Greece

14.4 % China

9.9 % Japan

The top 10 owner countries hold 67 % of the fleet.

Whose flag it flies

Share of world deadweight, by flag

17.4 % Liberia

15.2 % Panama

12.5 % Marshall Islands

About 70 % of capacity flies a flag other than the owner's. **Portugal's Madeira register (MAR) is 11th, 1.5 %, and grew 21.8 % in 2024.**

Who builds it

Share of gross tonnage delivered in 2024

54.6 % China

28.0 % Rep. of Korea

12.6 % Japan

Europe builds the high-value niche: cruise ships, ferries and special ships.

Who recycles it

Share of gross tonnage scrapped in 2024

43.2 % Bangladesh

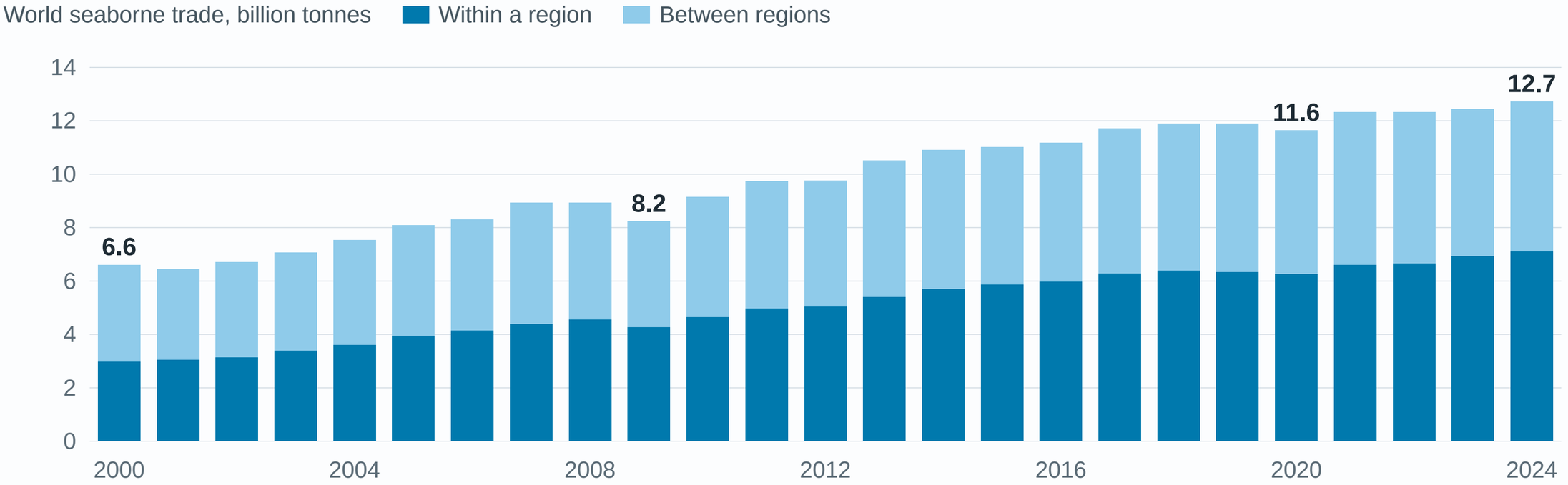
33.9 % India

7.4 % Türkiye

Only 0.25 % of the fleet was scrapped in 2024. The Hong Kong Convention is in force since June 2025.

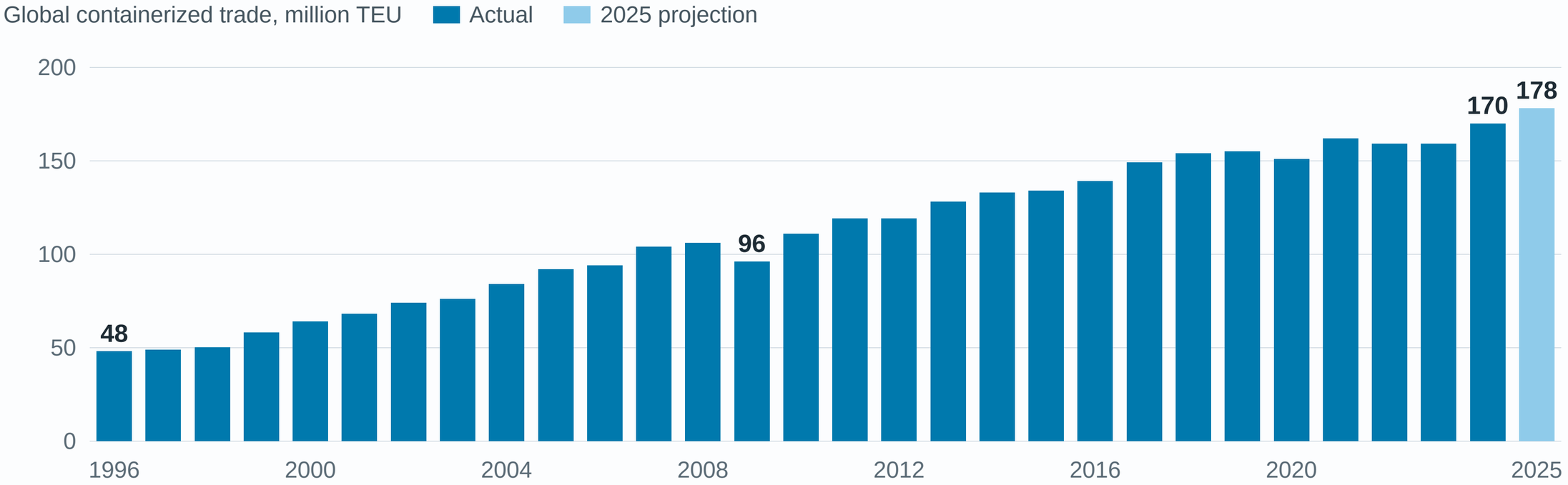
The fuel transition is in the orderbook, not yet in the fleet: **53 %** of the tonnage on order can run on alternative fuels, against **8 %** of the fleet in service (May 2025). UNCTAD (2025), Review of Maritime Transport, Ch. II §7, p. 62, after Clarksons Research.

12.7 Billion Tonnes a Year Go by Sea



Seaborne trade almost doubled in 24 years. In 2024 the fleet carried about **35 million tonnes every day**, falling only in 2009 (financial crisis) and 2020 (COVID-19).

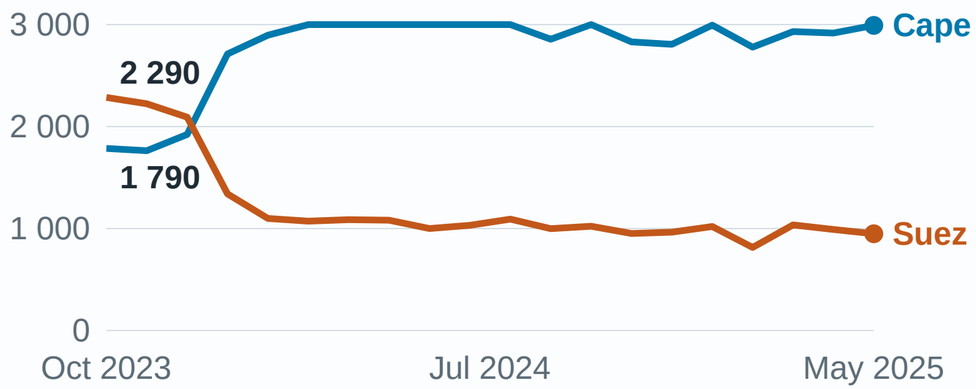
Containers: 170 Million TEU a Year



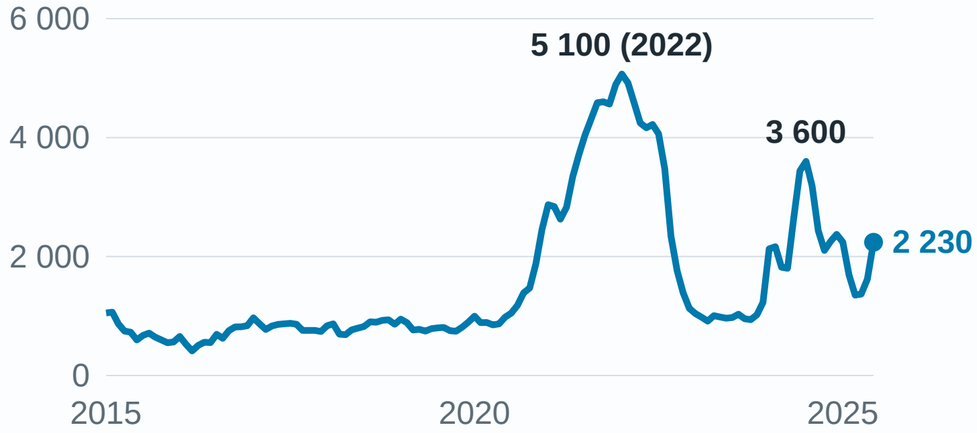
Container trade grew about **3.5 times** since 1996. The world container fleet that carries it: about 6,000 ships with 30 million TEU of capacity (April 2025).

Stormy Seas: Longer Routes, Volatile Rates

Shipping lanes under strain: ships a month



Volatile freight: container spot rate, USD per TEU



+0.5 %
seaborne trade expected in 2025, after +2.2 % in 2024

+5.9 %
ton-miles in 2024, a record: distance grew faster than volume

+5 %
greenhouse gas emissions from shipping in 2024

80 %+
of world merchandise trade moves by sea

Where the Baseline Ships Sit

Criterion	A · Azores Express	B · Azores Feeder
StatCode 5 type	Container Ship (Fully Cellular), gearless	Container Ship (Fully Cellular), own crane
Size	1,300 TEU · feeder class	130 TEU · small feeder
Operating area	Open sea: Lisbon–Azores, weekly	Short sea: inter-island loops
Hull	Conventional container ship	Hull of the group’s choice
Machinery	Dual-fuel engine, single shaft, PTI	Two electric motors; generator sets for transit
Energy source	HFO and LNG	Swappable battery containers

> Same StatCode 5 ship type (Cargo Carrying › Dry Cargo/Passenger › Container), very different ships: 0.2 Ship Design Project, slides 4 and 5, has the owner’s requirements of each. **Next: the ship types of this course, then Activity 1.1.**

Next session, Ship types 1.2: what limits the ship, its deadweight, its volume or one of its dimensions.

Merchant Ships: the Focus of the Course

1 Dry bulk carriers

Handysize to Capesize bulk carriers, ore carriers, self-unloaders

W1.4

2 General and special cargo

General cargo and multi-purpose ships, heavy-lift and semi-submersible ships, reefers, livestock carriers

W2.3

3 Container ships

Feeder to ultra-large, gearless and geared

W1.3

4 Ro-Ro and vehicle carriers

Ro-Ro, Con-Ro, pure car and truck carriers (PCC, PCTC)

W2.4

5 Oil and chemical tankers

Crude (Aframax to VLCC), shuttle, product, chemical and bunker tankers

W3.1 · W3.2

6 Gas carriers

LPG, LNG, ammonia and CO2 carriers; liquid hydrogen next

W3.3

7 Passenger ships and yachts

Ro-Pax and fast ferries, cruise ships, yachts

W3.4 · W4.1

Our ships: the reference ship is in group 5; the Azores Express and Feeder are in group 3.

Energy at Sea, Naval Vessels and the Rest

8 Offshore oil and gas

Drilling units (semi-submersible, jack-up, drill-ship) · production and storage (FPSO, FSO, FLNG, spar, TLP) · support vessels (PSV, AHTS) · construction (pipe-layers, crane vessels, DSV)

W4.2 · W4.3 · W5.1 · W5.2

9 Offshore wind

Installation vessels (WTIV, heavy-lift jack-ups) · floating foundations and cable-layers · service vessels (SOV, CTV)

W5.3 · W5.4 · W6.1

10 Naval and government vessels

Frigates, corvettes, destroyers · patrol, coastguard and mine countermeasure vessels, uncrewed vessels · amphibious, replenishment and support ships, submarines

W6.2 · W7.1 · W7.2

11 Tugs, dredgers and service vessels

Tugs, icebreakers, dredgers, pilot vessels, buoy tenders, oil-recovery, fire-fighting, research and survey, training ships

Today only

12 Fishing vessels

Trawlers, purse seiners, longliners, fish carriers

Today only

In total: 9 sessions on merchant ships, 4 on offshore oil and gas, 3 on offshore wind, 3 on naval vessels.

ACTIVITY 1.1 · INDIVIDUAL · 15 MIN · NOT GRADED

Find Each Ship in StatCode 5

Open the StatCode 5 Explorer (course site, W1.1). For each ship, write its five levels, from family to detailed type.

#	Ship	Your answer
1	The reference ship: a 37,600 t oil/chemical tanker	Family › … › detailed type
2	A ferry carrying cars and passengers between the Azores islands	Family › … › detailed type
3	A passenger ferry that only sails on a river	Family › … › detailed type
4	A service operation vessel for an offshore wind farm	Family › … › detailed type
5	A ship of your choice	Family › … › detailed type

Then: which ship surprised you, and at which level do ships 2 and 3 part ways?